

Lecture 27: Final Project Work Session

Models of Language and Conversation 

PSYC 51.07: Models of Language and Communication

Today's Agenda

Where We Are

The Home Stretch

This is it—the final week of the course! Your final projects are due soon, and presentations are coming up.

This week's schedule:

- **Today (Monday):** Project work session, debugging support, presentation prep
- **Wednesday:** Final project presentations (Part 1)
- **Friday:** Final project presentations (Part 2)
- **End of Friday:** All final materials due (11:59 PM Eastern)

Important

Today is your **last structured work time** before presentations. Use it wisely!

Final Project Requirements Review

What you need to submit (all due Friday EOD):

1. **Code Implementation:** Jupyter notebook that runs in Google Colab
 - Well-documented, reproducible, polished
 - Includes all experiments and results
2. **Video Presentation:** 10-12 minute video (YouTube link)
 - Cover motivation, approach, results, discussion
 - All team members participate
3. **Written Writeup:** 2-5 page report
 - Intro, approach, results, discussion, references
 - Think mini research paper

Presentation Format

Video Component:

- **Length:** 10-12 minutes
- **Format:** Pre-recorded video
- **Platform:** YouTube (unlisted OK)
- **Content:**
 - Motivation (why?)
 - Approach (how?)
 - Results (what?)
 - Discussion (so what?)

In-Class Component:

What Makes a Great Presentation? ★

Content:

- **Clear motivation:** Why does this problem matter?
- **Technical depth:** Show you understand the methods
- **Honest results:** Good and bad—what worked, what didn't?
- **Critical analysis:** Limitations, future work, broader implications

Presentation quality:

- **Clear visuals:** Readable slides, effective figures
- **Good pacing:** Not too fast, not too slow
- **Engaging delivery:** Energy, clarity, enthusiasm
- **Team coordination:** Smooth transitions, balanced participation

Presentation Structure Template

Suggested 10-12 minute structure:

1. Title & Introduction (1 min)

- Project title, team members, one-sentence summary

2. Motivation & Background (2 min)

- The problem, why it matters, research questions

3. Approach & Methods (3-4 min)

- Models, datasets, techniques, implementation details

4. Results (3-4 min)

- Key findings, metrics, visualizations, demos

5. Discussion & Conclusions (1-2 min)

- Limitations, future work, takeaways, broader impact

Finishing Your Project: Technical Checklist

Code & Implementation:

- ☐ Notebook runs completely in fresh Colab session (no errors!)
- ☐ All required packages installed with `!pip install` commands
- ☐ Dataset downloads automated (no manual file uploads)
- ☐ Random seeds set for reproducibility
- ☐ Code is well-commented and organized
- ☐ Markdown cells explain what's happening and why

Results & Analysis:

- ☐ All experiments completed and documented
- ☐ Visualizations clear and informative
- ☐ Metrics and headlines included

Testing Reproducibility

Critical Step: Test in Clean Environment

Before submitting, open a **brand new** Colab session and run your notebook from top to bottom. This is what graders will do!

Common reproducibility issues:

1. **Missing installations:** Did you forget a `!pip install` ?
2. **File paths:** Are you referencing files that only exist on your machine?
3. **API keys:** Did you hard-code keys? (Use environment variables or instructions)
4. **Cell order:** Does notebook work when run sequentially?
5. **Out-of-order execution:** Did you run cells out of order during development?
6. **Runtime requirements:** Does it need a GPU? Does it exceed Colab limits?

Use "Runtime → Restart and run all" frequently during development!

Code Quality Matters

Good code is:

- **Readable:** Clear variable names, logical organization
- **Documented:** Comments, docstrings, markdown explanations
- **Modular:** Functions for repeated tasks, not copy-paste code
- **Efficient:** Reasonable runtime, not wasteful

Example of good documentation:

```
1 def compute_embeddings(texts, model_name="sentence-transformers/all-MiniLM-L6-v2"):  
2     """  
3     Compute semantic embeddings for a list of texts.  
4  
5     Args:  
6         texts (list): List of text strings to embed  
7         model_name (str): The name of the model to use for embedding. Default is "sentence-transformers/all-MiniLM-L6-v2".  
8     """  
9     # Compute embeddings using the specified model  
10    embeddings = SentenceTransformer(model_name).embed(texts)  
11    return embeddings
```

Common Pitfalls to Avoid ⚠

Technical pitfalls:

- ❌ **Scope creep:** Trying to do too much at the last minute
- ❌ **No baselines:** Results without comparison are hard to interpret
- ❌ **Cherry-picking:** Only showing best results, hiding failures
- ❌ **Ignoring errors:** Sweeping bugs under the rug instead of fixing
- ❌ **Last-minute testing:** Discovering critical bugs right before deadline

Presentation pitfalls:

- ❌ **Too much detail:** Getting lost in implementation minutiae
- ❌ **No motivation:** Jumping straight to methods without context
- ❌ **Unclear visuals:** Tiny fonts, cluttered slides, confusing plots
- ❌ **Going over time:** Practice and time yourself!

When Things Don't Work

Research Reality

Not everything will work perfectly. That's **completely normal** in research!

If experiments fail or results are negative:

- **Don't panic!** Negative results are still valuable
- **Analyze why:** What went wrong? What did you learn?
- **Be honest:** Discuss failures and limitations openly
- **Show process:** Document what you tried and why
- **Suggest improvements:** What would you do differently?

Projects are graded on process, rigor, and critical thinking—not just positive results. A thoughtful analysis of why something didn't work can be just as valuable as showing success!

Time Management: The Final 72 Hours

Recommended priorities:

1. **Priority 1 (Critical):** Make code run and reproduce core results

- Fix any breaking bugs
- Ensure reproducibility
- Complete essential experiments

2. **Priority 2 (High):** Create presentation video

- Draft slides
- Record and edit video
- Upload to YouTube

3. **Priority 3 (Medium):** Polish notebook

- Add documentation

Grading Breakdown

How your project will be evaluated:

Results \	Analysis	20%
Presentation (Video + In-Class)	15%	
Written Writeup	10%	
Proposal	10%	
Teamwork \	Collaboration	5%

Focus most of your remaining time on **implementation** and **presentation**—they make up 55% of the grade!

What We're Looking For ★

Excellent projects demonstrate:

- **Technical sophistication:** Mastery of course concepts
- **Rigor:** Well-designed experiments, appropriate evaluation
- **Critical thinking:** Thoughtful analysis of results and limitations
- **Clarity:** Clear communication in code, writing, and presentation
- **Reproducibility:** Everything runs and can be verified
- **Insight:** Novel findings or meaningful contributions

Think about it!

Would you be proud to show this project in a job interview or grad school application?

Would you be comfortable presenting it at a research conference? That's the bar we're aiming for.

Today's Work Session

Use today for:

- **Debugging:** Fix critical bugs, resolve issues
- **Completing experiments:** Finish any remaining analyses
- **Creating presentation:** Draft slides, record video
- **Testing reproducibility:** Run notebook in clean Colab
- **Peer feedback:** Show work to other teams, get suggestions
- **Getting help:** Ask questions, troubleshoot with instructor

Office Hours Available

I'll be available all day for questions, debugging help, and feedback. Use this time!

Work together in your team, but also talk to other teams. Fresh perspectives often reveal issues you've been missing!

Getting Unstuck

If you're stuck on a technical issue:

1. **Read the error message carefully:** What exactly is breaking?
2. **Google the error:** Someone has likely encountered it before
3. **Check documentation:** Are you using the API correctly?
4. **Simplify:** Create minimal example that reproduces the issue
5. **Use GenAI tools:** Claude, ChatGPT can help debug
6. **Ask for help:** Discord, peers, instructor

If you're stuck on project direction:

1. **Review your proposal:** What was your original plan?
2. **Prioritize:** What's essential vs. nice-to-have?
2. **Scope down:** Better to do less well than more poorly

Key Takeaways

1. **Prioritize ruthlessly:** Focus on what matters most
2. **Test reproducibility:** Run in clean Colab before submitting
3. **Be honest about limitations:** Negative results are OK
4. **Practice your presentation:** Time it, refine it, polish it
5. **Use available support:** Office hours, peers, Discord
6. **Start with working code:** Polish can come later
7. **Document as you go:** Don't leave it all for the last minute

Remember

This is the culmination of everything you've learned. From ELIZA to GPT, from pattern matching to transformers—you've built an incredible foundation. Now show us what you can do with it!

Final Submission Checklist

Before you submit Friday night, verify:

- ☐ Code runs completely in fresh Colab (tested!)
- ☐ Video uploaded to YouTube, link included
- ☐ Writeup is 2-5 pages with all required sections
- ☐ All team members listed on all deliverables
- ☐ References properly cited
- ☐ Presentation ready for Wednesday or Friday
- ☐ All files submitted to Canvas

Submit early if possible! Don't wait until 11:59 PM—technical issues happen.

You've Got This! 💪

You've learned an incredible amount this quarter.

You started with simple string manipulation and pattern matching...

And now you're building sophisticated LLM systems!

Trust your skills. Use your resources. Do great work.

We can't wait to see your presentations! 🎉

Now: Let's get to work! 🚀